Chapter 9. An Introduction to Elliptic Functions

Doubly periodic functions, Liouville's theorems, the Weierstrass $\wp$ function, Eisenstein series

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Overview

Elliptic functions grew out of *elliptic integrals* $\int R(x, \sqrt{P(x)})\,dx$, R rational, P of degree three or four (arc length of an ellipse or lemniscate), and their double periodicity (cf. Section 4). Jacobi built the theory on theta functions; we sketch Weierstrass's simpler approach via $\wp$, reaching the *Eisenstein series* and divisor functions. Theta functions follow in Chapter 10.

1 Elliptic functions

Let f be meromorphic on $\mathbb{C}$ and **doubly periodic**: $f(z+\omega_1) = f(z+\omega_2) = f(z)$ for non-zero ω_1, ω_2. If $\omega_2/\omega_1 \in \mathbb{R}$, f is simply periodic or constant, so assume ω_1, ω_2 are $\mathbb{R}$-independent. *Normalization*: with $\tau = \omega_2/\omega_1$, interchanging if needed (τ, $1/\tau$ have imaginary parts of opposite signs), $\mathrm{Im}(\tau) > 0$; f has periods ω_1, ω_2 iff $F(z) = f(\omega_1 z)$ has periods $1, \tau$. So assume periods 1 and τ, $\mathrm{Im}(\tau) > 0$. Then

$$f(z + n + m\tau) = f(z) \qquad \text{for all integers } n, m \text{ and all } z \in \mathbb{C}, \tag{1}$$

and we consider the **lattice** $\Lambda = \{n+m\tau : n, m \in \mathbb{Z}\}$ **generated** by 1 and τ, and the **fundamental parallelogram**

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau,\ 0 \le a < 1,\ 0 \le b < 1\}.$$

$z \sim w$ (**congruent modulo** Λ) if $z - w \in \Lambda$; then $f(z) = f(w)$ by (1). Every z is congruent to a unique point of P_0: with $z = a+b\tau$ ($a, b \in \mathbb{R}$) and n, m the integer parts of a, b, $w = z-n-m\tau \in P_0$; if $a + b\tau \sim a' + b'\tau$ in P_0, then $a - a', b - b' \in \mathbb{Z} \cap (-1, 1) = \{0\}$. A **period parallelogram** is a translate $P = P_0 + h$; applied to $z - h$, the same holds for P. Moreover

$$\mathbb{C} = \bigcup_{n,m\in\mathbb{Z}} (n + m\tau + P_0), \tag{2}$$

a disjoint union.

Proposition 1.1

Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ. Then:
(i) Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram.
(ii) Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram.
(iii) The lattice Λ provides a disjoint covering of the complex plane, in the sense of (2).
(iv) The function f is completely determined by its values in any period parallelogram.

1.1 Liouville's theorems

Theorem 1.2

An entire doubly periodic function is constant.

Proof. It is determined by its values on P_0, whose closure is compact; so it is bounded on $\mathbb{C}$, hence constant by Liouville's theorem. ■

A non-constant doubly periodic meromorphic function is an **elliptic function**; it has finitely many zeros and poles in a period parallelogram, counted with multiplicity.

Theorem 1.3

The total number of poles of an elliptic function in P_0 is always ≥ 2.

So f cannot have just one simple pole (a single double pole is allowed).

Proof. Suppose first f has no poles on ∂P_0. By the residue theorem $\int_{\partial P_0} f(z)\,dz = 2\pi i \sum \operatorname{res} f$, and the integral is 0 by periodicity:

$$\int_{\partial P_0} f = \int_0^1 f + \int_1^{1+\tau} f + \int_{1+\tau}^{\tau} f + \int_{\tau}^0 f,$$

and opposite sides cancel; e.g.

$$\int_0^1 f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz = \int_0^1 f(z)\,dz + \int_1^0 f(s+\tau)\,ds = \int_0^1 f(z)\,dz - \int_0^1 f(s)\,ds = 0.$$

So $\sum \operatorname{res} f = 0$, and f has at least two poles in P_0. With a pole on ∂P_0, use $P = h + P_0$ for a small h. ■

The total number of poles (with multiplicities) of an elliptic function is its **order**.

Theorem 1.4

Every elliptic function of order m has m zeros in P_0.

Proof. With no zeros or poles on ∂P_0, $\int_{\partial P_0} \frac{f'}{f} = 2\pi i(N_z - N_p)$ vanishes by periodicity; otherwise use a translate. ■

Consequently, for every $c \in \mathbb{C}$, $f(z) = c$ has as many solutions in P_0 as the order of f ($f - c$ is elliptic with the same poles).

1.2 The Weierstrass $\wp$ function

An elliptic function of order two. We want double poles exactly at the lattice points. For one period, $\sum_n \frac{1}{z+n}$ is 1-periodic with poles at the integers but not absolutely convergent; summing symmetrically,

$$F(z) = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \Big[\frac{1}{z+n} + \frac{1}{z-n}\Big],$$

the bracket being $O(1/n^2)$; $F(z) = \pi \cot \pi z$ (Chapter 5). Equivalently $\frac{1}{z} + \sum_{n\neq 0} \big[\frac{1}{z+n} - \frac{1}{n}\big]$, absolutely convergent, since pairing $\pm n$ gives $\frac{1}{z+n} + \frac{1}{z-n}$. Mimicking this, as $\sum_{\omega\in\Lambda} \frac{1}{(z+\omega)^2}$ is not absolutely convergent, with $\Lambda^* = \Lambda - \{0\}$ consider

$$\frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big], \qquad \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2 - 2z\omega}{(z+\omega)^2\omega^2} = O\Big(\frac{1}{|\omega|^3}\Big) \text{ as } |\omega| \to \infty.$$

Lemma 1.5

The two series

$$\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} \qquad \text{and} \qquad \sum_{n+m\tau\in\Lambda^*} \frac{1}{|n+m\tau|^r}$$

converge if $r > 2$.

(By the Note at the end of Chapter 7 the order of summation does not matter; we sum in m, then n.)

Proof. For $n \neq 0$, by integral comparison ($1/k^r \le \int_{k-1}^{k} dx/x^r$),

$$\sum_{m\in\mathbb{Z}} \frac{1}{(|n|+|m|)^r} = \frac{1}{|n|^r} + 2\sum_{k\ge|n|+1} \frac{1}{k^r} \le \frac{1}{|n|^r} + 2\int_{|n|}^{\infty} \frac{dx}{x^r} \le \frac{1}{|n|^r} + \frac{C}{|n|^{r-1}},$$

so for $r>2$, $\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} = \sum_{|m|\neq 0} \frac{1}{|m|^r} + \sum_{|n|\neq 0}\sum_m \frac{1}{(|n|+|m|)^r} < \infty$. For the second series it suffices that $|n|+|m| \le c|n+m\tau|$. Write $A \approx B$ if $A \lesssim B \lesssim A$ ($x \lesssim y$: $x \le ay$ for some $a>0$). For $A, B > 0$, $(A^2+B^2)^{1/2} \approx A+B$ (square; and $A, B \le (A^2+B^2)^{1/2}$). With $\tau = s+it$, $t>0$,

$$|n+m\tau| = [(n+ms)^2 + (mt)^2]^{1/2} \approx |n+ms| + |mt| \approx |n+ms| + |m|,$$

and $|n+ms|+|m| \approx |n|+|m|$, considering separately $|n| \ge 2|m||s|$ and $|n| < 2|m||s|$. ∎

Remark

The proof shows that for $r>2$ the series $\sum |n+m\tau|^{-r}$ converges uniformly in every half-plane $\mathrm{Im}(\tau) \ge \delta > 0$. For $r=2$ it diverges.

The **Weierstrass $\wp$ function** is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big] = \frac{1}{z^2} + \sum_{(n,m)\neq(0,0)} \Big[\frac{1}{(z+n+m\tau)^2} - \frac{1}{(n+m\tau)^2}\Big].$$

It is meromorphic with double poles at the lattice points: for $|z| < R$ the terms with $|\omega| > 2R$ are $O(1/|\omega|^3)$ uniformly, so their sum is holomorphic (Lemma 1.5), and the finitely many others give the double poles.

Theorem 1.6

The function $\wp$ is an elliptic function that has periods 1 and τ, and double poles at the lattice points.

Proof. It remains to prove periodicity. Differentiating termwise,

$$\wp'(z) = -2\sum_{n,m\in\mathbb{Z}} \frac{1}{(z+n+m\tau)^3},$$

which converges absolutely for $z \notin \Lambda$ ($r=3$ in Lemma 1.5) and has periods 1, τ. So $\wp(z+1) = \wp(z)+a$, $\wp(z+\tau) = \wp(z)+b$. $\wp$ is even ($\omega \mapsto -\omega$), so $z=-1/2$, $z=-\tau/2$ give $a=b=0$. ∎

Properties of $\wp$. $\wp$ is even, so $\wp'$ is odd; with periodicity,

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\Big(\frac{1+\tau}{2}\Big) = 0,$$

e.g. $\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2+1) = -\wp'(1/2)$. Since $\wp'$ is elliptic of order 3, the **half-periods** $1/2$, $\tau/2$, $(1+\tau)/2$ are its only roots in P_0, each of multiplicity 1. Let

$$\wp(1/2) = e_1, \qquad \wp(\tau/2) = e_2, \qquad \wp\Big(\frac{1+\tau}{2}\Big) = e_3.$$

$\wp(z) = e_1$ has a double root at $1/2$ and, $\wp$ having order 2, no others in P_0; likewise e_2, e_3. So e_1, e_2, e_3 are distinct.

Theorem 1.7

The function $(\wp')^2$ is the cubic polynomial in $\wp$

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. $F = (\wp - e_1)(\wp - e_2)(\wp - e_3)$ and $(\wp')^2$ both have, in P_0, only double roots at $1/2$, $\tau/2$, $(1+\tau)/2$ and poles of order 6 at the lattice points, so $(\wp')^2/F$ is constant; from $\wp = z^{-2} + \cdots$, $\wp' = -2z^{-3} + \cdots$ it is 4. ■

$\wp$ is universal:

Theorem 1.8

Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.

Lemma 1.9

Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.

Proof. A zero or pole of the even F at 0 has even order, so replacing F by $\wp^m F$ we may assume none at the lattice points. Recall $\wp(z) - \wp(a)$ has a double zero if a is a half-period and zeros a, $-a$ otherwise. Zeros of F come in pairs $a, -a$ ($a \sim -a$ iff a is a half-period, then of even order). So if $a_1, -a_1, \ldots, a_m, -a_m$ are all zeros in a period parallelogram (with multiplicities, half for each of $a_j, -a_j$ at a half-period), $[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]$ has exactly the roots of F. Similarly with the poles $b_1, -b_1, \ldots, b_m, -b_m$,

$$G(z) = \frac{[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)] \cdots [\wp(z) - \wp(b_m)]}$$

is periodic with the same zeros and poles as F. So F/G is holomorphic and doubly periodic, hence constant. ■

Proof of Theorem 1.8. Write $f = f_{\text{even}} + f_{\text{odd}}$ with $f_{\text{even}}(z) = \frac{f(z)+f(-z)}{2}$, $f_{\text{odd}}(z) = \frac{f(z)-f(-z)}{2}$. Since $\wp'$ is odd, $f_{\text{odd}}/\wp'$ is even; apply the lemma to f_{even} and $f_{\text{odd}}/\wp'$. ■

2 The modular character of elliptic functions and Eisenstein series

Now write $\wp_\tau$. As $1, \tau$ and $1, \tau + 1$ generate the same lattice, $\wp_\tau = \wp_{\tau+1}$; $-1/\tau$ essentially interchanges the periods, and $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$. So we consider the **modular group** generated by $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$ on $\mathbb{H}$.

2.1 Eisenstein series

The **Eisenstein series of order** k is

$$E_k(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n + m\tau)^k} = \sum_{\omega \in \Lambda^*} \frac{1}{\omega^k},$$

for integers $k \geq 3$ and $\operatorname{Im}(\tau) > 0$.

Theorem 2.1

Eisenstein series have the following properties:
(i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane.
(ii) $E_k(\tau) = 0$ if k is odd.
(iii) $E_k(\tau)$ satisfies the following transformation relations: $E_k(\tau+1) = E_k(\tau)$ and $E_k(\tau) = \tau^{-k}E_k(-1/\tau)$.

Property (iii) is the **modular character** of the Eisenstein series.

Proof. (i) By Lemma 1.5 and the remark after it, the series converges absolutely and uniformly in every half-plane $\mathrm{Im}(\tau) \geq \delta > 0$ for $k \geq 3$.

(ii) Replacing (n, m) by $(-n, -m)$ shows $E_k = -E_k$ for odd k.

(iii) $n + m(\tau+1) = (n+m) + m\tau$, and we may replace $n+m$ by n. Also $(n + m(-1/\tau))^k = \tau^{-k}(n\tau - m)^k$, and rearranging, replacing $(-m, n)$ by (n, m), gives the second relation. ∎

(Because of (ii), some authors define the Eisenstein series as $\sum 1/(n+m\tau)^{2k}$, possibly with a constant factor.)

Theorem 2.2

For z near 0, we have

$$\wp(z) = \frac{1}{z^2} + 3E_4 z^2 + 5E_6 z^4 + \cdots = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k}.$$

Proof. Replacing ω by $-\omega$, $\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*}\left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right]$. Differentiating the geometric series, $\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty}(\ell+1)w^\ell$ for $|w| < 1$, so for small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2}\sum_{\ell=0}^{\infty}(\ell+1)\Big(\frac{z}{\omega}\Big)^\ell = \frac{1}{\omega^2} + \sum_{\ell=1}^{\infty}(\ell+1)\frac{z^\ell}{\omega^{\ell+2}}.$$

Hence

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)\Big(\sum_{\omega\in\Lambda^*}\frac{1}{\omega^{\ell+2}}\Big)z^\ell = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)E_{\ell+2}z^\ell = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k},$$

since $E_{\ell+2} = 0$ for odd ℓ. ∎

Hence near 0,

$$\wp'(z) = \frac{-2}{z^3} + 6E_4 z + 20E_6 z^3 + \cdots,$$
$$(\wp'(z))^2 = \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \qquad (\wp(z))^3 = \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots.$$

So $(\wp')^2 - 4\wp^3 + 60E_4\wp + 140E_6$ is holomorphic near 0 and vanishes there; being doubly periodic, it is constant (Theorem 1.2), hence 0:

Corollary 2.3

If $g_2 = 60E_4$ and $g_3 = 140E_6$, then $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

This is another version of Theorem 1.7, and expresses the symmetric functions of the e_j via Eisenstein series.

2.2 Eisenstein series and divisor functions

Consider the Fourier expansion of the periodic function $E_k(\tau)$; equivalently, the Laurent expansion in $z = e^{2\pi i\tau}$.

Lemma 2.4

If $k \geq 2$ and $\mathrm{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell \tau}.$$

Proof. One may apply Poisson summation to $f(z) = 1/(z+\tau)^k$. Alternatively, it suffices to treat $k = 2$, the other cases following by termwise differentiation. Differentiating $\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau$ (Chapter 5) gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

With $w = e^{2\pi i\tau}$ ($|w| < 1$), $\sin^2 \pi\tau = -\frac{(e^{i\pi\tau} - e^{-i\pi\tau})^2}{4} = -\frac{(1-w)^2}{4w}$, so $\frac{\pi^2}{\sin^2 \pi\tau} = -4\pi^2 \frac{w}{(1-w)^2} = (-2\pi i)^2 \sum_{r=1}^{\infty} r w^r$, using $\sum_{r \geq 1} r w^r = \frac{w}{(1-w)^2}$. ∎

The **divisor function** $\sigma_\ell(r) = \sum_{d|r} d^\ell$ is the sum of the ℓth powers of the divisors of r.

Theorem 2.5

If $k \geq 4$ is even, and $\mathrm{Im}(\tau) > 0$, then

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. $\sigma_{k-1}(r) \leq \sum_{d \leq r} d^{k-1} \leq r^k$; if $\mathrm{Im}(\tau) \geq t_0 > 0$, $|e^{2\pi i r\tau}| \leq e^{-2\pi r t_0}$, so the series converges absolutely, by comparison with $\sum r^k e^{-2\pi r t_0}$. Using the definition, k even, and the lemma with τ replaced by $m\tau$,

$$\begin{aligned} E_k(\tau) &= \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} = 2\zeta(k) + 2 \sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2\frac{(-2\pi i)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m \tau} = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i r \tau}, \end{aligned}$$

collecting the terms with $\ell m = r$. ∎

For the "forbidden" case $k = 2$, $\sum_{(n,m) \neq (0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but define

$$F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}, \qquad (n,m) \neq (0,0),$$

summed in the indicated order (first n, then m).

Corollary 2.6

The double sum defining F converges in the indicated order. We have

$$F(\tau) = 2\zeta(2) - 8\pi^2 \sum_{r=1}^{\infty} \sigma(r) e^{2\pi i r \tau},$$

where $\sigma(r) = \sum_{d|r} d$ is the sum of the divisors of r.

(Theorem 2.5's argument applies with $k = 2$.) $F(-1/\tau)\tau^{-2} \neq F(\tau)$: summing first in m then in n changes the value. This series is nevertheless crucial for the four-squares theorem (Chapter 10).

3 Selected exercises

Motivation. Exercises from Stein–Shakarchi (numbers kept).

Exercise 9.1 (Real ratio of periods)

Let f be meromorphic on $\mathbb{C}$ with non-zero periods ω_1, ω_2 such that $\omega_2/\omega_1 \in \mathbb{R}$.
(a) If $\omega_2/\omega_1 = p/q$ with p, q relatively prime integers, $q \geq 1$, show that f has the periods ω_1, ω_2 if and only if f has the single period $\omega_0 = \omega_1/q$.
(b) If ω_2/ω_1 is irrational, show that f is constant.

Solution. If ω, ω' are periods, so are $k\omega$, $k \in \mathbb{Z}$ (induction, and $f(z-\omega) = f((z-\omega)+\omega)$) and $\omega + \omega'$; so all integer combinations are periods.

(a)

Step 1. If ω_0 is a period, then $\omega_1 = q\omega_0$ and $\omega_2 = \frac{p}{q}\omega_1 = p\omega_0$ are integer multiples of it, hence periods.

Step 2. Conversely, since $\gcd(p, q) = 1$ there are integers m, n with $mq + np = 1$. Then

$$m\omega_1 + n\omega_2 = \omega_1\Big(m + n\frac{p}{q}\Big) = \frac{\omega_1}{q}(mq + np) = \omega_0,$$

an integer combination of ω_1, ω_2, hence a period.

(b) Let $\alpha = \omega_2/\omega_1 \notin \mathbb{Q}$ and $D = \{m + n\alpha : m, n \in \mathbb{Z}\}$.

Step 1. (D is dense in $\mathbb{R}$.) Of the $N+1$ fractional parts of $0, \alpha, \ldots, N\alpha$, two (of $j\alpha$, $j'\alpha$) lie in one $[i/N, (i+1)/N)$, so $\delta = (j - j')\alpha - \ell$ has $0 < |\delta| < 1/N$ (α irrational). The multiples $k\delta \in D$ meet every interval of length $1/N$.

Step 2. (Many small periods.) For $t = m + n\alpha \in D$, $t\omega_1 = m\omega_1 + n\omega_2$ is a period. By Step 1 choose $t_k \in D \setminus \{0\}$ with $t_k \to 0$.

Step 3. (Conclusion.) The pole set S is discrete and closed, so $U = \mathbb{C} \setminus S$ is open and connected. For $z_0 \in U$, $z_k = z_0 + t_k\omega_1 \to z_0$, $z_k \neq z_0$, $f(z_k) = f(z_0)$ (Step 2), so $f \equiv f(z_0)$ on U; then f has no poles, $S = \varnothing$, and f is constant. ∎

Exercise 9.2 (Zeros minus poles)

Let f be elliptic with periods $1, \tau$, lattice Λ, zeros $a_1, \ldots, a_r$ and poles $b_1, \ldots, b_r$ in P_0 with multiplicity (equally many, Theorem 1.4). Show that

$$a_1 + \cdots + a_r - b_1 - \cdots - b_r \in \Lambda.$$

Solution.

Step 1. (Choice of a parallelogram.) For each of the countably many zeros and poles w, the h with $w \in \partial(h + P_0)$ form $w - \partial P_0$, of area zero; so some h puts none on ∂P, $P = h + P_0$.

Step 2. (Reduction to P.) By Proposition 1.1 there is a bijection $\phi : P_0 \to P$ with $\phi(z) - z \in \Lambda$, preserving orders of zeros and poles. So the zeros and poles in P are $a_i' = \phi(a_i)$, $b_j' = \phi(b_j)$, and

$$\textstyle\sum_i a_i - \sum_j b_j = \sum_i a_i' - \sum_j b_j' - \lambda \qquad \text{for some } \lambda \in \Lambda.$$

So it suffices to show $\sum_i a_i' - \sum_j b_j' \in \Lambda$.

Step 3. (Residues.) At a zero a of order k, zf'/f has residue ka; at a pole b of order k, $-kb$; elsewhere near $\overline{P}$ it is holomorphic. By the residue theorem,

$$\frac{1}{2\pi i}\int_{\partial P} z\frac{f'(z)}{f(z)}\,dz = \sum_i a_i' - \sum_j b_j'.$$

Step 4. (Pairing opposite sides.) Let $g = f'/f$, which has periods 1 and τ. The boundary runs $h \to h+1 \to h+1+\tau \to h+\tau \to h$. Substituting $z \mapsto z+\tau$ in the third side and $z \mapsto z+1$ in the second,

$$\int_h^{h+1} zg\,dz + \int_{h+1+\tau}^{h+\tau} zg\,dz = \int_h^{h+1} [z-(z+\tau)]g(z)\,dz = -\tau\int_h^{h+1} g(z)\,dz,$$
$$\int_{h+1}^{h+1+\tau} zg\,dz + \int_{h+\tau}^{h} zg\,dz = \int_h^{h+\tau} [(z+1)-z]g(z)\,dz = \int_h^{h+\tau} g(z)\,dz.$$

Step 5. (Integers.) The curve $\gamma(t) = f(h+t)$, $0 \le t \le 1$, is closed because $f(h+1) = f(h)$, and avoids 0. Hence

$$\int_h^{h+1} \frac{f'(z)}{f(z)}\,dz = \int_\gamma \frac{dw}{w} = 2\pi i\, n$$

with $n \in \mathbb{Z}$ the winding number of γ about 0. In the same way $\int_h^{h+\tau} g = 2\pi i\,\ell$ with $\ell \in \mathbb{Z}$.

Step 6. Combining Steps 3–5, $\sum_i a_i' - \sum_j b_j' = \ell - n\tau \in \Lambda$. ∎

Exercise 9.3 (The exponent 2 is critical)

(a) Show that $\sum_{\omega\in\Lambda^*}|\omega|^{-2}$ diverges for every $\tau \in \mathbb{H}$.
(b) Show that, as $R \to \infty$,

$$\sum_{1\le n^2+m^2\le R^2} \frac{1}{n^2+m^2} = 2\pi\log R + O(1).$$

Solution.

(b) Write $\omega = (n,m) \in \mathbb{Z}^2$, $f(x) = |x|^{-2}$ on $\mathbb{R}^2 \setminus \{0\}$, $c = \sqrt{2}/2$, and let Q_ω be the closed unit square centered at ω. Take $R \ge 4$.

Step 1. (Local comparison.) For $|\omega| \ge 2$, $x \in Q_\omega$, every $y \in [\omega, x] \subset Q_\omega$ has $|y| \ge |\omega|/2$ and $|\nabla f(y)| = 2|y|^{-3} \le 16|\omega|^{-3}$; the segment has length $\le c$, so

$$|f(x) - f(\omega)| \le 8\sqrt{2}\,|\omega|^{-3}.$$

Step 2. (Sum versus integral.) The six terms with $1 \le |\omega| < 2$ ($(\pm1,0)$, $(0,\pm1)$, $(\pm1,\pm1)$) total $4 + 4\cdot\frac{1}{2} = 6$; let $S_1(R)$ be the sum over $2 \le |\omega| \le R$ and U_R the union of the unit squares Q_ω there (disjoint interiors). By Step 1,

$$\Big|S_1(R) - \int_{U_R} f\,dx\Big| \le 8\sqrt{2}\sum_{|\omega|\ge 2}|\omega|^{-3} =: C < \infty,$$

the series converging by Lemma 1.5 with $r = 3$, $\tau = i$.

Step 3. (Annuli.) Let $A(a,b)=\{a\le|x|\le b\}$. If $2+c\le|x|\le R-c$, then x lies in some Q_ω, and $|x-\omega|\le c$ gives $2\le|\omega|\le R$; so $A(2+c,R-c)\subset U_R$. Conversely, $x\in Q_\omega$ with $2\le|\omega|\le R$ has $2-c\le|x|\le R+c$, so $U_R\subset A(2-c,R+c)$. In polar coordinates $\int_{A(a,b)}f=\int_0^{2\pi}\int_a^b r^{-1}\,dr\,d\theta=2\pi\log(b/a)$, and $f>0$, so

$$2\pi\log\frac{R-c}{2+c}\le\int_{U_R}f\,dx\le 2\pi\log\frac{R+c}{2-c}.$$

Step 4. (Conclusion.) For $R\ge 4$, $|\log(1\pm c/R)|\le\log 2$, so both bounds equal $2\pi\log R+O(1)$. With Step 2 the full sum is $6+2\pi\log R+O(1)=2\pi\log R+O(1)$.

(a) With $\tau\in\mathbb{H}$ and $\omega=n+m\tau\neq 0$,

$$|n+m\tau|\le|n|+|m||\tau|\le(1+|\tau|)(|n|+|m|)\le\sqrt{2}(1+|\tau|)(n^2+m^2)^{1/2}.$$

Hence $|\omega|^{-2}\ge\frac{1}{2(1+|\tau|)^2}(n^2+m^2)^{-1}$, and the partial sums over $n^2+m^2\le R^2$ are at least a fixed multiple of $2\pi\log R+O(1)\to\infty$ by (b). So the series diverges. ∎

Exercise 9.7 ($\zeta(2)$ and $\zeta(4)$ from lattice sums)

Using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-2}=\pi^2/\sin^2(\pi\tau)$ at $\tau=1/2$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^2}=\frac{\pi^2}{8},\qquad\zeta(2)=\frac{\pi^2}{6},$$

and, using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-4}$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^4}=\frac{\pi^4}{96},\qquad\zeta(4)=\frac{\pi^4}{90}.$$

Solution.

Step 1. (The identities on $\mathbb{C}\setminus\mathbb{Z}$.) By Chapter 5, $\pi\cot\pi z=\lim_N\sum_{|n|\le N}\frac{1}{z+n}$ for $z\notin\mathbb{Z}$, uniformly on compact subsets of $\mathbb{C}\setminus\mathbb{Z}$. Differentiating termwise (Weierstrass), as in the proof of Lemma 2.4,

$$\sum_{n\in\mathbb{Z}}\frac{1}{(z+n)^2}=\frac{\pi^2}{\sin^2\pi z}=:g(z),\qquad z\in\mathbb{C}\setminus\mathbb{Z},$$

and the left side converges absolutely and locally uniformly, so it may be differentiated twice more termwise:

$$\sum_{n\in\mathbb{Z}}\frac{6}{(z+n)^4}=g''(z).$$

Step 2. (Rewriting at $z=1/2$.) $(n+\frac12)^{-k}=2^k(2n+1)^{-k}$, and as n runs over $\mathbb{Z}$, $2n+1$ runs over the odd integers, each odd $m\ge1$ arising twice ($\pm m$). So for even k,

$$\sum_{n\in\mathbb{Z}}\frac{1}{(n+\frac12)^k}=2^{k+1}\sum_{m\ge1\text{ odd}}\frac{1}{m^k}.$$

Step 3. ($k=2$.) $g(1/2)=\pi^2$, so $\sum_{m\text{ odd}}m^{-2}=\pi^2/8$. Splitting $\zeta(2)$ into odd and even terms, $\zeta(2)=\frac{\pi^2}{8}+\frac14\zeta(2)$, hence $\zeta(2)=\pi^2/6$.

Step 4. ($k=4$.) With $u=\pi z$,

$$g'(z)=-2\pi^3\csc^2u\cot u,\qquad g''(z)=2\pi^4(2\csc^2u\cot^2u+\csc^4u).$$

At $z=1/2$, $\csc u=1$ and $\cot u=0$, so $g''(1/2)=2\pi^4$ and $\sum_n(n+\frac12)^{-4}=\pi^4/3$. By Step 2, $\sum_{m\text{ odd}}m^{-4}=\frac{\pi^4}{3\cdot32}=\frac{\pi^4}{96}$. Then $\zeta(4)=\frac{\pi^4}{96}+\frac{1}{16}\zeta(4)$, so $\zeta(4)=\frac{16}{15}\cdot\frac{\pi^4}{96}=\frac{\pi^4}{90}$. ∎